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Multi temperature generator and engine shared oil

Abstract

A fluid delivery system is provided including a reservoir including oil, a hydraulic pump in fluid communication with the reservoir, and a heat exchanger system. The heat exchanger system is in fluid communication with the hydraulic pump, a generator, and an engine gearbox. The hydraulic pump is configured to deliver the oil to the heat exchanger system. The heat exchanger system is configured to provide the oil according to first temperature criteria to the generator and provide the oil according to second temperature criteria to the engine gearbox.

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Background/Summary

BACKGROUND

(1) The present disclosure relates to heat exchangers and, in particular, to oil heat exchangers in aircraft applications.

BACKGROUND

(2) Heat exchangers can be used in various applications, including but not limited to aerospace, refrigeration, air conditioning, space heating, electricity generation, and chemical processing applications. For example, heat exchangers can be utilized within an aircraft to transfer heat from one fluid to another. Some aircraft may use air-based or fuel-based oil coolers to extract thermal energy from oil to maintain oil within target operational temperatures. In some cases, structures associated with heat exchangers may add complexity, expense, and weight to product designs, which may be undesirable with respect to cost-sensitive, weight-sensitive, and/or space-limited applications associated with an aircraft.

BRIEF DESCRIPTION

(3) According to an aspect of the disclosure, a fluid delivery system is provided including: a reservoir including oil; a hydraulic pump in fluid communication with the reservoir; and a heat exchanger system in fluid communication with the hydraulic pump, a generator, and an engine gearbox, wherein: the hydraulic pump is configured to deliver the oil to the heat exchanger system; and the heat exchanger system is configured to provide the oil according to first temperature criteria to the generator and provide the oil according to second temperature criteria to the engine gearbox.

(4) In some embodiments, the generator and the engine gearbox are associated with a gas turbine engine of a vehicle.

(5) In any one or combination of the embodiments disclosed herein, the generator and the engine gearbox are associated with an auxiliary power unit (APU) of a vehicle.

(6) In any one or combination of the embodiments disclosed herein, the heat exchanger system includes: a first heat exchanger configured to provide the oil according to the first temperature criteria to the generator; and a second heat exchanger configured to provide the oil according to the second temperature criteria to the engine gearbox.

(7) In any one or combination of the embodiments disclosed herein, the fluid delivery system is configured to direct a cooling medium to the first heat exchanger or the second heat exchanger based on respective priorities associated with the first heat exchanger and the second heat exchanger.

(8) In any one or combination of the embodiments disclosed herein, the heat exchanger system includes: a heat exchanger configured to provide the oil according to the first temperature criteria to the generator and provide the oil according to the second temperature criteria to the engine gearbox.

(9) In any one or combination of the embodiments disclosed herein, the heat exchanger system further includes: a scavenge pump in fluid communication with the generator, the engine gearbox, and the reservoir, wherein the scavenge pump is configured to: collect at least a portion of the oil from at least one of the generator and the engine gearbox; and provide at least the portion of the oil to the reservoir.

(10) In any one or combination of the embodiments disclosed herein, the heat exchanger system further includes at least one of: a first sensor device in thermal communication with an inlet of the heat exchanger system; a second sensor device in thermal communication with a first outlet of the heat exchanger system; a third sensor device in thermal communication with a second outlet of the heat exchanger system; a fourth sensor device in thermal communication with an outlet of the engine gearbox; and a fifth sensor device in thermal communication with an outlet of the generator, wherein the fluid delivery system is configured to maintain or adjust a temperature of the oil as received at the inlet of the heat exchanger system based on temperature information provided by at least one of the first sensor device, the second sensor device, the third sensor device, the fourth sensor device, and the fifth sensor device.

(11) In any one or combination of the embodiments disclosed herein, the heat exchanger system further includes: a sensor device in thermal communication with at least one of an outlet of the heat exchanger system and an inlet of the generator; and control circuitry coupled to the sensor device and the generator, wherein the control circuitry is configured to condition generation of electric power by the generator, based on temperature information provided by the sensor device.

(12) According to an aspect of the disclosure, a gas turbine engine system is provided including: a gas turbine engine; a generator; an engine gearbox configured to operatively connect the gas turbine engine to the generator; and a fluid delivery system. The fluid delivery system includes: a reservoir including oil; a hydraulic pump in fluid communication with the reservoir; and a heat exchanger system in fluid communication with the hydraulic pump, the generator, and the engine gearbox, wherein: the hydraulic pump is configured to deliver the oil to the heat exchanger system;

and the heat exchanger system is configured to provide the oil according to first temperature criteria to the generator and provide the oil according to second temperature criteria to the engine gearbox.

(13) In some embodiments, the heat exchanger system includes: a first heat exchanger configured to provide the oil according to the first temperature criteria to the generator; and a second heat exchanger configured to provide the oil according to the second temperature criteria to the engine gearbox.

(14) In any one or combination of the embodiments disclosed herein, the fluid delivery system is configured to direct a cooling medium to the first heat exchanger or the second heat exchanger based on respective priorities associated with the first heat exchanger and the second heat exchanger.

(15) In any one or combination of the embodiments disclosed herein, the heat exchanger system includes: a heat exchanger configured to provide the oil according to the first temperature criteria to the generator and provide the oil according to the second temperature criteria to the engine gearbox.

(16) In any one or combination of the embodiments disclosed herein, the gas turbine engine further includes: a scavenge pump in fluid communication with the generator, the engine gearbox, and the reservoir, wherein the scavenge pump is configured to: collect at least a portion of the oil from at least one of the generator and the engine gearbox; and provide at least the portion of the oil to the reservoir.

(17) In any one or combination of the embodiments disclosed herein, the gas turbine engine further includes at least one of: a first sensor device in thermal communication with an inlet of the heat exchanger system; a second sensor device in thermal communication with a first outlet of the heat exchanger system; a third sensor device in thermal communication with a second outlet of the heat exchanger system; a fourth sensor device in thermal communication with an outlet of the engine gearbox; and a fifth sensor device in thermal communication with an outlet of the generator, wherein the fluid delivery system is configured to maintain or adjust a temperature of the oil as received at the inlet of the heat exchanger system based on temperature information provided by at least one of the first sensor device, the second sensor device, the third sensor device, the fourth sensor device, and the fifth sensor device.

(18) In any one or combination of the embodiments disclosed herein, the gas turbine engine further includes: a sensor device in thermal communication with at least one of an outlet of the heat exchanger system and an inlet of the generator; and control circuitry coupled to the sensor device and the generator, wherein the control circuitry is configured to condition generation of electric power by the generator, based on temperature information provided by the sensor device.

(19) According to an aspect of the disclosure, a method of delivering a fluid to a generator and an engine gearbox of a gas turbine engine system is provided, the method including: determining, by a fluid delivery system, temperature information of oil received at an inlet of a heat exchanger system of the fluid delivery system; and maintaining or adjusting, by the fluid delivery system, a temperature of the oil based on the temperature information, wherein maintaining or adjusting the temperature of the oil includes: providing the oil to the generator according to first temperature criteria; and providing the oil to the engine gearbox according to second temperature criteria.

(20) In some embodiments, the method further includes: collecting, by a scavenge pump of the fluid delivery system, at least a portion of the oil from at least one of the generator and the engine gearbox; and providing at least the portion of the oil to a reservoir of the fluid delivery system.

(21) In any one or combination of the embodiments disclosed herein, the method further includes: determining, by the fluid delivery system, at least one of: second temperature information of the oil at a first outlet of the heat exchanger system; third temperature information of the oil at a second outlet of the heat exchanger system; fourth temperature information of the oil at an outlet of the engine gearbox; and fifth temperature information of the oil at an outlet of the generator, wherein

maintaining or adjusting the temperature of the oil received at the inlet of the heat exchanger system is further based on at least one of the second temperature information, the third temperature information, the fourth temperature information, and the fifth temperature information.

(22) In any one or combination of the embodiments disclosed herein, the method further includes: determining, by a sensor device, second temperature information of the oil at an outlet of the heat exchanger system; providing, by the fluid delivery system, a control signal to the generator based on the second temperature information, wherein the control signal is associated with conditioning generation of electric power by the generator.

(23) Additional features and advantages are realized through the techniques of the present disclosure. Other embodiments and aspects of the disclosure are described in detail herein and are considered a part of the claimed technical concept. For a better understanding of the disclosure with the advantages and the features, refer to the description and to the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of this disclosure, reference is now made to the following brief description, taken in connection with the accompanying drawings and detailed description, wherein like reference numerals represent like parts.

(2) FIG. 1 is a schematic illustration of an integrated oil system in accordance with one or more embodiments of the present disclosure.

(3) FIG. 2 illustrates an example flowchart in accordance with one or more embodiments of the present disclosure.

DETAILED DESCRIPTION

(4) A detailed description of one or more embodiments of the disclosed apparatus and method are presented herein by way of exemplification and not limitation with reference to the Figures.

(5) According to one or more embodiments of the present disclosure, a shared oil system is described which is capable of providing oil at multiple temperature outputs. In some embodiments, the shared oil system is a common oil system for an aircraft, in which the oil system includes a heat exchanger system for providing oil of different temperatures to an engine gearbox of the aircraft and a generator of the aircraft. In an example, the heat exchanger system includes a main engine heat exchanger for providing oil according to first temperature criteria (e.g., a first temperature, a first temperature range, or the like) to the engine gearbox. In some embodiments, the heat exchanger system includes an additional heat exchanger for providing oil according to second temperature criteria (e.g., a second temperature, a second temperature range, or the like) to the generator.

(6) As will be described herein, example aspects of the shared oil system support a reduction in weight, cost, instrumentation, volume, and quantity of components compared to other techniques for providing oil of differing temperatures within an aircraft. In some aspects, example embodiments provide a shared oil system having a reduced complexity (e.g., shared oil simplicity) while being capable of providing oil according to different temperature criteria. For example, the shared oil system is capable of providing oil according to relatively higher oil temperatures for the engine gearbox and providing oil according to relatively lower temperatures for the generator (e.g., using cooler oil may extend the life of the generator).

(7) FIG. 1 is a schematic illustration of a vehicle, e.g., an aircraft **100**, that can incorporate various embodiments of the present disclosure. The aircraft **100** includes an integrated oil system **101**. The integrated oil system **101** is described herein as being utilized by the aircraft **100**; however, the integrated oil system **101** is not limited thereto. Aspects of the present disclosure support utilizing the integrated oil system **101** in other various systems including, but not limited to, land-based

vehicles and naval vessels. The integrated oil system **101** is capable of providing fluid (e.g., oil) to components of the aircraft **100**. The integrated oil system **101** may be referred to herein as a fluid delivery system.

(8) In accordance with one or more embodiments of the present disclosure, the integrated oil system **101** is capable of providing or delivering oil at different temperatures to a generator **135** and an engine gearbox **137** of the aircraft **100**.

(9) In some embodiments, the generator **135** may be capable of generating electrical power for the aircraft **100**. For example, the generator **135** may be an integrated drive generator (IDG) or variable frequency generator (VFG). The generator **135** may provide electrical power to components of the aircraft **100** via an electrical system (not illustrated) of the aircraft **100**. However, aspects of the present disclosure are not limited thereto, and the generator **135** may be a generator, a motor/generator, or a motor. In some aspects, the generator **135** (e.g., as a motor or motor/generator) may be capable of adding mechanical energy back to the gas turbine engine **102**.

(10) The engine gearbox **137** may be coupled to or integrated with a gas turbine engine **102** of the aircraft **100**. In some embodiments, the engine gearbox **137** may be a main gearbox or an accessory gearbox capable of distributing the high-speed rotational movement generated by the gas turbine engine **102** to components included in the aircraft **100**. In some examples, the engine gearbox **137** may be a mechanically driven gearbox for driving systems of the aircraft **100** such as, for example, fuel pumps, scavenge pumps (e.g., scavenge pump **140**), electrical generators (e.g., generator **135**), hydraulic pumps (e.g., hydraulic pump **110**), and the like. For example, the engine gearbox **137** may transfer power from the main shaft of the gas turbine engine **102** to the generator **135** for electrical power. It is to be understood that the engine gearbox **137** may include, but is not limited to, any or all of the following: an engine accessory gear box and an engine main gear box (e.g., such as used to reduce or increase the fan or compressor speed or speed to accessory gearbox and bearing compartments used to support the engine rotating fan, compressor, and or turbine shafts).

(11) The example aspects of the integrated oil system **101** are not limited to a gas turbine engine **102**. For example, in some other embodiments, the generator **135** and the engine gearbox **137** described herein may be associated with an auxiliary power unit (APU) (not illustrated) of the aircraft **100**. For example, the APU may include the generator **135** and the engine gearbox **137**.

(12) The integrated oil system **101** includes a reservoir **105** (also referred to herein as an oil tank or a lubricant tank), a hydraulic pump **110**, a system filter **115**, a main oil pressure sensor **120**, a heat exchanger system **123**, sensor devices **130** (e.g., temperature sensors), generator **135**, engine gearbox **137**, and scavenge pump **140**.

(13) The integrated oil system **101** may store solid fluids (e.g., oil) in reservoir **105**. In some examples, the reservoir **105** may include a temperature transducer to monitor the temperature of fluid stored in the reservoir **105**, and the reservoir **105** may include one or more vents configured to vent the reservoir **105** to atmospheric pressure. Further, the reservoir **105** may include various components to manage and/or service the oil volume therein including, but not limited to, fluid fill and spill ports, a fluid level sight glass, and a fluid drain plug.

(14) The integrated oil system **101** includes a pump system capable of drawing in fluid (e.g., oil **121**) contained in reservoir **105** and delivering the fluid to the system filter **115**, and further, to the heat exchanger system **123**. The pump system may include hydraulic pump **110** (also referred to herein as an oil pump) configured to deliver the fluid to the system filter **115** and/or heat exchanger system **123**.

(15) In some embodiments, the integrated oil system **101** may include a bypass valve (not illustrated) via which the integrated oil system **101** may bypass the system filter **115** (e.g., and thereby bypass oil **121** around the system filter **115**) for cases in which the system filter **115** is blocked. Accordingly, for example, the integrated oil system **101** may provide the oil **121** to heat exchanger system **123**, with or without filtering of the oil **121** by the system filter **115**.

(16) In some other embodiments, the integrated oil system **101** may include a bypass valve (not

illustrated) via which the integrated oil system **101** may bypass the heat exchanger system **123** (e.g., and thereby bypass oil **121** around the heat exchanger system **123**). For example, the integrated oil system **101** may bypass the heat exchanger system **123** based on the temperature of the oil **121**. For example, the integrated oil system **101** may include a sensor device **130-a** thermally coupled to the inlet **122** of the heat exchanger system **123** and configured to measure the temperature of the oil **121** at the inlet **122** of the heat exchanger system **123**. In an example, for cases in which the temperature of the oil **121** is equal to or substantially equal to a target temperature for a target component (e.g., generator **135**, engine gearbox **137**), the integrated oil system **101** may provide the oil **121** to the target component, without heating or cooling of the oil **121** by the heat exchanger system **123**.

(17) Main oil pressure sensor is capable sensing pressure (e.g., increases or decreases in pressure) across the hydraulic pump **110**.

(18) The heat exchanger system **123** is capable of providing oil **127-a** and oil **127-b** of different respective temperatures to generator **135** and engine gearbox **137**. In some embodiments, the heat exchanger system **123** may be as a single heat exchanger capable of providing oil **127-a** and oil **127-b**. In some other embodiments, the heat exchanger system **123** may be implemented as a heat exchanger **125-a** and a heat exchanger **125-b** respectively capable of providing oil **127-a** and oil **127-b** of different respective temperatures. It is to be understood that in the examples described herein, oil **127-a** and oil **127-b** may be the same as oil **121**, but at different respective temperatures.

(19) The heat exchanger system **123** may support modifying the temperature of oil **121** and outputting oil **127-a** and oil **127-b** of different respective temperatures. In an example, the heat exchanger system **123** may heat or cool the oil **121** to a first temperature or a first temperature range associated with lubricating the generator **135**, and the heat exchanger system **123** may heat or cool the oil **121** to a second temperature or a second temperature range associated with lubricating the engine gearbox **137**. In another example, the heat exchanger **125-a** may cool the oil **121** to the first temperature or the first temperature range, and the heat exchanger **125-b** may cool the oil **121** to the second temperature or the second temperature range.

(20) In an example, the heat exchanger **125-a** may be a relatively large heat exchanger capable of cooling all available oil **121** from the reservoir **105** to the first temperature or the first temperature range associated with lubricating the generator **135** (e.g., thereby providing oil **127-a**). In the example, the heat exchanger **125-b** may be a relatively smaller heat exchanger capable of further cooling the oil **127-a** to the second temperature or the second temperature range associated with lubricating the engine gearbox **137** (e.g., thereby providing oil **127-b**). In some aspects, the heat exchanger system **123** may further include one or more other one or more heat exchangers **125** (not illustrated) for further cooling the oil **127-a** or the oil **127-b**.

(21) Accordingly, for example, the integrated oil system **101** may support providing any suitable quantity of streams of lubricant (e.g., oil **127-a**, oil **127-b**, and the like) according to respective target temperatures. For example, the integrated oil system **101** may provide, from the reservoir **105**, oil according to one or more target temperatures (e.g., for generator **135**, for engine gearbox **137**, for other components to which oil of a given temperature is to be provided).

(22) In some embodiments, the target temperature of the oil **121** for the heat exchanger **125-a** and/or the heat exchanger **125-b** may range from about 70 degrees Celsius to about 120 degrees Celsius. In an example, the first temperature range for the oil **121** and the heat exchanger **125-a** may be about 70 degrees Celsius to about 120 degrees Celsius. In some examples, the second temperature range for the oil **121** and the heat exchanger **125-b** may be about 70 degrees Celsius to about 120 degrees Celsius.

(23) In some embodiments, the integrated oil system **101** may give the generator **135** priority on the incoming cooling medium (e.g., fuel, air, or the like). For example, the integrated oil system **101** is configured to direct the cooling medium to the heat exchanger **125-a** or the heat exchanger **125-b** based on respective priorities associated with the heat exchanger **125-a** and the heat

exchanger **125-b**. In an example, the integrated oil system **101** may first provide the cooling medium to the heat exchanger **125-a** associated with the generator **135**, followed by providing the cooling medium to the heat exchanger **125-b** associated with the engine gearbox **137**. In an example, the temperature of the cooling medium may increase after being used at the heat exchanger **125-a**, and the integrated oil system **101** may further provide the same cooling medium (in some cases, having an increased temperature) to the heat exchanger **125-b**.

(24) The heat exchangers described herein (e.g., heat exchanger system **123**, heat exchanger **125-a**, heat exchanger **125-b**) may include any heat exchanger supportive of maintaining or adjusting the temperature of the oil **121** (also referred to herein as the oil flow of oil **121**). For example, the heat exchangers described herein may be capable of heating, cooling, or maintaining the temperature of the oil **121** in association with providing corresponding oil **127-a** to the generator **135** and/or providing corresponding oil **127-b** to the engine gearbox **137**. Aspects described herein with reference to heat exchanger system **123** may be similarly applied to heat exchanger **125-a** and heat exchanger **125-b**.

(25) In some embodiments, the heat exchanger system **123** includes an air cooled oil cooler (not illustrated). In some other embodiments the heat exchanger system **123** includes a fuel cooled oil cooler (not illustrated). In some other embodiments, the heat exchanger system **123** may include an air cooled oil cooler and a fuel cooled oil cooler in a single unit.

(26) In an embodiment, the heat exchanger system **123** may support cooling portions of the oil **121** by the air cooled oil cooler, while cooling other portions of the oil **121** by the fuel cooled oil cooler. The air cooled oil cooler may be disposed to receive an air flow (e.g., a fan air flow associated with the gas turbine engine **102**, an external air flow from outside the aircraft **100**, or the like). In certain embodiments, the air cooled oil cooler supports cross flow heat transfer. In certain embodiments, the fuel cooled oil cooler supports cross flow, counter flow, or parallel flow heat transfer.

(27) In certain embodiments, the heat exchanger system **123** may include one or more paths via which the oil **121** can partially or fully bypass the heat exchanger system **123**. Advantageously, the heat exchanger system **123** supports features for selectively cooling the oil **121** through various cooling paths such that the oil **121** is cooled to or remains within a target temperature or a target temperature range.

(28) The heat exchanger system **123** includes an inlet **122** for receiving oil **121**, a flow outlet **126-a** for providing oil **127-a**, and a flow outlet **126-b** for providing oil **127-b**. The heat exchanger system **123** may include one or more proportioning devices (e.g., one or more proportioning valves **124**) for selectively directing portions of the oil **121** to a given component (e.g., heat exchanger **125-a**, heat exchanger **125-b**, air cooled oil cooler, fuel cooled oil cooler, or the like) of the heat exchanger system **123**.

(29) Non-limiting examples of heat exchanger system **123** include laminate-type heat exchangers, plate type, plate-shell type, tube type, and tube-shell type heat exchangers supportive of maintaining or adjusting (e.g., cooling, heating, or the like) the temperature of the oil **121**.

(30) The integrated oil system **101** may include sensor devices **130** (e.g., temperature sensors) capable of measuring temperatures of the oil **121**, the oil **127-a**, the oil **127-b**, oil **136** which leaks or drains out of the generator **135**, and/or oil **138** which leaks or drains out of the engine gearbox **137**. In some embodiments, the sensor devices **130** can provide continuous and/or semi-continuous feedback of temperature information any of oil **121** (e.g., as received at the inlet **122** of heat exchanger system **123**, as stored at reservoir **105**, or the like), oil **127-a**, oil **127-b**, oil **136**, and oil **138** to aircraft computers or any other suitable control system (including control circuitry and/or processing circuitry) associated with the aircraft **100**. In some examples, the temperature information may include measured temperatures, temperature patterns, temperature statistical data, or the like of any of oil **121** (e.g., as received at the inlet **122** of heat exchanger system **123**, as stored at reservoir **105**, or the like), oil **127-a**, oil **127-b**, oil **136**, and oil **138**.

(31) For example, based on temperature information (e.g., measured temperatures) provided by one

or more of the sensor devices **130**, the integrated oil system **101** may control one or more proportioning valves **124** in association with maintaining and/or adjusting the temperatures of oil (e.g., oil **127-a**, oil **127-b**) to be provided to a component of the aircraft **100**. In some examples, the integrated oil system **101** may throttle one or more one or more proportioning valves **124** in association with maintaining and/or adjusting the temperatures of oil described herein.

(32) In some aspects, based on the temperature information (e.g., measured temperatures) provided by the sensor devices **130**, the integrated oil system **101** may evaluate the performance of the heat exchanger system **123** (or heat exchanger **125-a** and/or heat exchanger **125-b**), and the integrated oil system **101** may maintain or adjust one or more parameters of the heat exchanger system **123**. Accordingly, for example, based on the temperature information provided by the sensor devices **130**, the integrated oil system **101** may regulate the temperatures of the oil **127-a** and/or the oil **127-b**.

(33) The scavenge pump **140** may be in fluid communication with the generator **135**, the engine gearbox **137**, and the reservoir **105**. In some embodiments, the scavenge pump **140** may be a positive displacement scavenge pump. The integrated oil system **101** may collect, by the scavenge pump **140**, leakage and drainage fluid from various portions of the integrated oil system **101** (e.g., from generator **135**, from engine gearbox **137**, a clutch control system (not illustrated), gear/bearing systems, or the like). In an example, the scavenge pump **140** may be driven by a rotating input shaft **112** coupled between the hydraulic pump **110** and the scavenge pump **140**.

(34) In some embodiments, the integrated oil system **101** may transfer collected oil from the scavenge pump **140** to the reservoir **105**. Additionally, or alternatively, the hydraulic pump **110** may draw collected oil from the scavenge pump **140**, provide the collected oil to the system filter **115**, and provide the filtered oil to the heat exchanger system **123**.

(35) The example embodiments described herein support providing oil of different temperatures to other components of the aircraft **100**, and it is to be understood that the techniques described herein are not limited to providing oil of different temperatures to the generator **135** and the engine gearbox **137**. For example, in one or more embodiments, the integrated oil system **101** may provide oil **127-a**, oil **127-b**, oil of another temperature as provided by heat exchanger system **123**, or a combined oil flow to an main bearings of a gas turbine engine **102** or to any other suitable component of the aircraft **100**.

(36) According to one or more additional and/or alternative embodiments supported by aspects of the present disclosure, the integrated oil system **101** may include a control loop (not illustrated) between the sensor device **130-a** and generator **135**. In some aspects, the control loop include control circuitry associated with providing power to electronic components of the aircraft **100**. In some aspects, the control loop may support the conditioning of electric power provided by generator **135**.

(37) The integrated oil system **101** may be capable of providing temperature information regarding individual circuits of the integrated oil system **101** to, for example, aircraft control circuitry of the aircraft **100**. In some aspects, based on the temperature information, the aircraft control circuitry may infer the health of components (e.g., generator **135**, engine gearbox **137**) of the integrated oil system **101** and aircraft **100**.

(38) For example, the integrated oil system **101** may include a control system (not illustrated), and the control system may determine the health of the generator **135** based on a comparison of the temperature of oil **127-a** at an inlet of the generator **135** and the temperature of oil **136** at an outlet of the generator **135**. In an example, for a condition in which the temperature of oil **136** at the outlet of the generator **135** rises, but the temperature of oil **127-a** at the inlet of the generator **135** remains constant, the generator control system may associate the condition as a failure condition within the generator **135**. Additionally, or alternatively, the control system may similarly determine the health of the engine gearbox **137** based on a comparison of the temperature of oil **127-b** at an inlet of the engine gearbox **137** and the temperature of oil **138** at an outlet of the **137**.

(39) According to one or more embodiments of the present disclosure, the reservoir **105**, the hydraulic pump **110**, the system filter **115**, the Component main oil pressure sensor, and the heat exchanger system **123** may be included in or integrated with the generator **135**. Accordingly, for example, the integrated oil system **101** is capable of providing oil **127-a** and oil **127-b** of different respective temperatures to generator **135** and engine gearbox **137** (and/or other components of the aircraft **100**) with a reduced number of components compared to some other systems, thereby achieving reduced complexity, expense, and weight. That is, for example, the integrated oil system **101** may be implemented without a separate reservoir (e.g., oil tank), hydraulic pump, system filter, main oil pressure sensor **120**, heat exchanger, and scavenge pump for providing oil **127-b** for the engine gearbox **137**. The integrated oil system **101** mitigates the implementation of redundant components and a redundant feedback loop for providing oil **127-b** for the engine gearbox **137**. In some aspects, the integrated oil system **101** described herein support an increase in oil compatibility or overlap, for example, as the same oil type may be provided by the integrated oil system **101** to both the generator **135** and the engine gearbox **137**.

(40) Accordingly, for example, aspects of the integrated oil system **101** described herein support a reduction in total maintenance costs (TMC) and direct maintenance costs (DMC) associated with maintaining the integrated oil system **101** for the aircraft **100**.

(41) FIG. 2 illustrates an example flowchart of a method **200** in accordance with one or more embodiments of the present disclosure. The method **200** may be implemented by the example aspects of an integrated oil system **101** described herein.

(42) At **205**, the method **200** includes determining, by a fluid delivery system, temperature information of oil received at an inlet of a heat exchanger system of the fluid delivery system.

(43) At **210**, the method **200** includes maintaining or adjusting, by the fluid delivery system, a temperature of the oil based on the temperature information.

(44) In some embodiments, maintaining or adjusting the temperature of the oil includes providing (at **215**) the oil to the generator according to first temperature criteria.

(45) In some embodiments, maintaining or adjusting the temperature of the oil includes providing (at **220**) the oil to the engine gearbox according to second temperature criteria.

(46) In some embodiments, the method **200** further includes: collecting, by a scavenge pump of the fluid delivery system, at least a portion of the oil from at least one of the generator and the engine gearbox; and providing at least the portion of the oil to a reservoir of the fluid delivery system.

(47) In any one or combination of the embodiments disclosed herein, the method **200** further includes: determining, by the fluid delivery system, at least one of: second temperature information of the oil at a first outlet of the heat exchanger system; third temperature information of the oil at a second outlet of the heat exchanger system; fourth temperature information of the oil at an outlet of the engine gearbox; and fifth temperature information of the oil at an outlet of the generator, wherein maintaining or adjusting the temperature of the oil received at the inlet of the heat exchanger system is further based on at least one of the second temperature information, the third temperature information, the fourth temperature information, and the fifth temperature information.

(48) In any one or combination of the embodiments disclosed herein, the method **200** further includes: determining, by a sensor device, second temperature information of the oil at an outlet of the heat exchanger system; providing, by the fluid delivery system, a control signal to the generator based on the second temperature information, wherein the control signal is associated with conditioning generation of electric power by the generator.

(49) In the descriptions of the flowcharts herein, the operations may be performed in a different order than the order shown, or the operations may be performed in different orders or at different times. Certain operations may also be left out of the flowcharts, one or more operations may be repeated, or other operations may be added to the flowcharts.

(50) The term “about” is intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application.

(51) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, element components, and/or groups thereof.

(52) While the present disclosure has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the present disclosure. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the essential scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this present disclosure, but that the present disclosure will include all embodiments falling within the scope of the claims.

(53) The corresponding structures, materials, acts and equivalents of all means or step plus function elements in the claims below are intended to include any structure, material, or act for performing the function in combination with other claimed elements as specifically claimed. The description of the present disclosure has been presented for purposes of illustration and description, but is not intended to be exhaustive or limited to the technical concepts in the form disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the disclosure. The embodiments were chosen and described in order to best explain the principles of the disclosure and the practical application and to enable others of ordinary skill in the art to understand the disclosure for various embodiments with various modifications as are suited to the particular use contemplated.

(54) While the various embodiments to the disclosure have been described, it will be understood that those skilled in the art, both now and in the future, may make various improvements and enhancements which fall within the scope of the claims which follow. These claims should be construed to maintain the proper protection for the disclosure first described.

Claims

1. A fluid delivery system comprising: a reservoir comprising oil; a hydraulic pump in fluid communication with the reservoir; a heat exchanger system in fluid communication with the hydraulic pump, a generator, and an engine gearbox; a control system; and a set of sensor devices configured to provide temperature information of the oil at one or more branches of the fluid delivery system, the one or more branches comprising an inlet of the generator and an outlet of the generator, wherein: the hydraulic pump is configured to deliver the oil to the heat exchanger system; the heat exchanger system is configured to provide the oil according to first temperature criteria to the generator and provide the oil according to second temperature criteria to the engine gearbox; and the control system is configured to determine a health of the generator based on: temperature information of the oil at the inlet of the generator; and temperature information of the oil at the outlet of the generator.

2. The fluid delivery system of claim 1, wherein the generator and the engine gearbox are associated with a gas turbine engine of a vehicle.

3. The fluid delivery system of claim 1, wherein the generator and the engine gearbox are associated with an auxiliary power unit (APU) of a vehicle.

4. The fluid delivery system of claim 1, wherein the heat exchanger system comprises: a first heat exchanger configured to provide the oil according to the first temperature criteria to the generator; and a second heat exchanger configured to provide the oil according to the second temperature

criteria to the engine gearbox.

5. The fluid delivery system of claim 4, wherein the fluid delivery system is configured to direct a cooling medium to the first heat exchanger or the second heat exchanger based on respective priorities associated with the first heat exchanger and the second heat exchanger.

6. The fluid delivery system of claim 1, wherein the heat exchanger system comprises: a heat exchanger configured to provide the oil according to the first temperature criteria to the generator and provide the oil according to the second temperature criteria to the engine gearbox.

7. The fluid delivery system of claim 1, further comprising: a scavenge pump in fluid communication with the generator, the engine gearbox, and the reservoir, wherein the scavenge pump is configured to: collect at least a portion of the oil from at least one of the generator and the engine gearbox; and provide at least the portion of the oil to the reservoir.

8. The fluid delivery system of claim 1, wherein the set of sensor devices comprises: a first sensor device in thermal communication with an inlet of the heat exchanger system and configured to provide first temperature information of the oil at an inlet of the heat exchanger system; a second sensor device in thermal communication with a first outlet of the heat exchanger system and configured to provide second temperature information of the oil at the first outlet of the heat exchanger system; a third sensor device in thermal communication with a second outlet of the heat exchanger system and configured to provide third temperature information of the oil at the second outlet of the heat exchanger system; a fourth sensor device in thermal communication with an outlet of the engine gearbox and configured to provide fourth temperature information of the oil at the outlet of the engine gearbox; and a fifth sensor device in thermal communication with an outlet of the generator and configured to provide the temperature information of the oil at the outlet of the generator, wherein the fluid delivery system is configured to: maintain or adjust a temperature of the oil as received at the inlet of the heat exchanger system based on the first temperature information of the oil at the inlet of the heat exchanger system, the second temperature information of the oil at the first outlet of the heat exchanger system, and the third temperature information of the oil at the second outlet of the heat exchanger system; maintain or adjust a temperature of the oil as received at the inlet of the engine gearbox based on the fourth temperature information of the oil at the outlet of the engine gearbox; and maintain or adjust a temperature of the oil as received at the inlet of the generator based on the temperature information of the oil at the outlet of the generator, wherein each of the first temperature information, the second temperature information, the third temperature information, the fourth temperature information, and the temperature information of the oil at the outlet of the generator comprises temperature patterns or temperature statistical data of the oil.

9. The fluid delivery system of claim 1, wherein: the set of sensor devices comprises a sensor device in thermal communication with at least one of an outlet of the heat exchanger system and an inlet of the generator; and the fluid delivery system further comprises control circuitry coupled to the sensor device and the generator, wherein the control circuitry is configured to condition generation of electric power by the generator, based on temperature information provided by the sensor device.

10. A gas turbine engine system comprising: a gas turbine engine; a generator; an engine gearbox configured to operatively connect the gas turbine engine to the generator; and a fluid delivery system comprising: a reservoir comprising oil; a hydraulic pump in fluid communication with the reservoir; a heat exchanger system in fluid communication with the hydraulic pump, the generator, and the engine gearbox; a control system; and a set of sensor devices configured to provide temperature information of the oil at one or more branches of the fluid delivery system, the one or more branches comprising an inlet to the generator and an outlet of the generator, wherein: the hydraulic pump is configured to deliver the oil to the heat exchanger system; the heat exchanger system is configured to provide the oil according to first temperature criteria to the generator and provide the oil according to second temperature criteria to the engine gearbox; and the control

system is configured to determine a health of the generator based on: temperature information of the oil at the inlet to the generator; and temperature information of the oil at the outlet of the generator.

11. The gas turbine engine of claim 10, wherein the heat exchanger system comprises: a first heat exchanger configured to provide the oil according to the first temperature criteria to the generator; and a second heat exchanger configured to provide the oil according to the second temperature criteria to the engine gearbox.

12. The gas turbine engine of claim 11, wherein the fluid delivery system is configured to direct a cooling medium to the first heat exchanger or the second heat exchanger based on respective priorities associated with the first heat exchanger and the second heat exchanger.

13. The gas turbine engine of claim 10, wherein the heat exchanger system comprises: a heat exchanger configured to provide the oil according to the first temperature criteria to the generator and provide the oil according to the second temperature criteria to the engine gearbox.

14. The gas turbine engine of claim 10, further comprising: a scavenge pump in fluid communication with the generator, the engine gearbox, and the reservoir, wherein the scavenge pump is configured to: collect at least a portion of the oil from at least one of the generator and the engine gearbox; and provide at least the portion of the oil to the reservoir.

15. The gas turbine engine of claim 10, wherein the set of sensor devices comprises: a first sensor device in thermal communication with an inlet of the heat exchanger system and configured to provide first temperature information of the oil at an inlet of the heat exchanger system; a second sensor device in thermal communication with a first outlet of the heat exchanger system and configured to provide second temperature information of the oil at the first outlet of the heat exchanger system; a third sensor device in thermal communication with a second outlet of the heat exchanger system and configured to provide third temperature information of the oil at the second outlet of the heat exchanger system; a fourth sensor device in thermal communication with an outlet of the engine gearbox and configured to provide fourth temperature information of the oil at the outlet of the engine gearbox; and a fifth sensor device in thermal communication with an outlet of the generator and configured to provide the temperature information of the oil at the outlet of the generator, wherein the fluid delivery system is configured to: maintain or adjust the temperature of the oil as received at the inlet of the heat exchanger system based on the first temperature information of the oil at the inlet of the heat exchanger system, the second temperature information of the oil at the first outlet of the heat exchanger system, and the third temperature information of the oil at the second outlet of the heat exchanger system; maintain or adjust a temperature of the oil as received at the inlet of the engine gearbox based on the fourth temperature information of the oil at the outlet of the engine gearbox; and maintain or adjust the temperature of the oil as received at the inlet of the generator based on the temperature information of the oil at the outlet of the generator, wherein each of the first temperature information, the second temperature information, the third temperature information, the fourth temperature information, and the temperature information of the oil at the outlet of the generator comprises temperature patterns or temperature statistical data of the oil.

16. The gas turbine engine of claim 10, wherein: the set of sensor devices comprises a sensor device in thermal communication with at least one of an outlet of the heat exchanger system and the inlet of the generator; and the fluid delivery system further comprises control circuitry coupled to the sensor device and the generator, wherein the control circuitry is configured to condition generation of electric power by the generator, based on temperature information provided by the sensor device.

17. A fluid delivery system comprising: a reservoir comprising oil; a hydraulic pump in fluid communication with the reservoir; a heat exchanger system in fluid communication with the hydraulic pump, a generator, and an engine gearbox; a control system; and a set of sensor devices comprising: a first sensor device in thermal communication with an inlet of the engine gearbox and

configured to provide first temperature information of the oil at the inlet of the engine gearbox; a second sensor device in thermal communication with an outlet of the engine gearbox and configured to provide second temperature information of the oil at the outlet of the engine gearbox, wherein: the hydraulic pump is configured to deliver the oil to the heat exchanger system; the heat exchanger system is configured to provide the oil according to first temperature criteria to the generator and provide the oil according to second temperature criteria to the engine gearbox; and the control system is configured to determine a health of the engine gearbox based on: the first temperature information of the oil at the inlet of the engine gearbox; and the second temperature information of the oil at the outlet of the engine gearbox.

18. The fluid delivery system of claim 17, wherein: the set of sensor devices further comprises: a third sensor device in thermal communication with an inlet of the heat exchanger system and configured to provide third temperature information of the oil at the inlet of the heat exchanger system; a fourth sensor device in thermal communication with a first outlet of the heat exchanger system and configured to provide fourth temperature information of the oil at the first outlet of the heat exchanger system; and a fifth sensor device in thermal communication with a second outlet of the heat exchanger system and configured to provide fifth temperature information of the oil at the second outlet of the heat exchanger system; and the fluid delivery system is configured to maintain or adjust a temperature of the oil as received at the inlet of the heat exchanger system further based on one or more of: the first temperature information of the oil at the inlet of the engine gearbox; the second temperature information of the oil at the outlet of the engine gearbox; the third temperature information of the oil at the inlet of the heat exchanger system; the fourth temperature information of the oil at the first outlet of the heat exchanger system; and the fifth temperature information of the oil at the second outlet of the heat exchanger system.
